

4 The mean resting heart rate varies between species of mammal.

The table shows the mean resting heart rate and mean lifespan for seven mammals.

Mammal	Mean resting heart rate / bpm	Mean lifespan / years
mouse	595	1
monkey	275	15
cat	120	17
giraffe	82	19
human	72	80
elephant	41	23
whale	15	29

(a) Describe **two** conclusions that can be drawn from these data.

(2)

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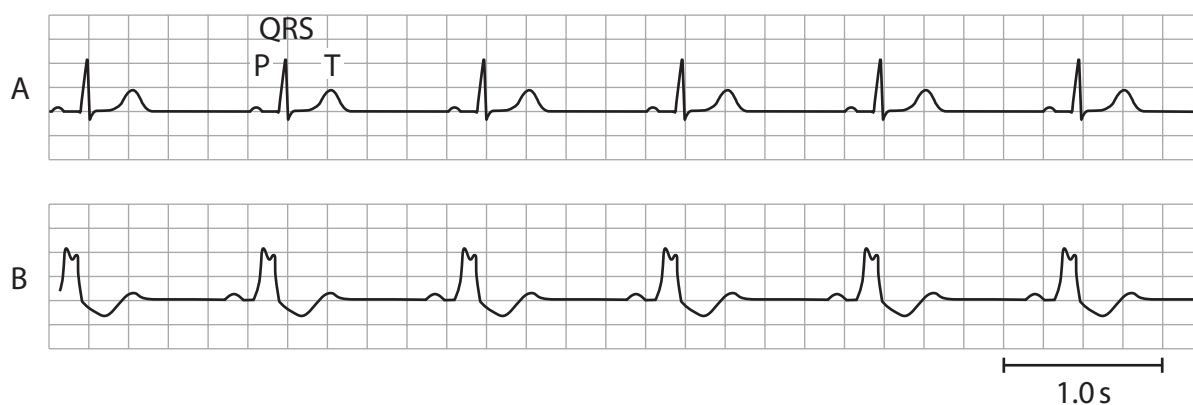
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(b) An electrocardiogram (ECG) is used to identify abnormal heart rhythms.

Trace A shows a normal heart rhythm and trace B shows an abnormal heart rhythm.



(i) Which is the heart rate shown in trace A?

(1)

- ☐ A 6 bpm
- ☐ B 48 bpm
- ☐ C 60 bpm
- ☐ D 75 bpm

(ii) Describe **two** differences shown between trace A and trace B.

(2)

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- (c) Cardiac muscle is unable to regenerate or repair itself. Mammalian heart muscle can beat only a certain number of times in a lifetime.

Humans have a mean resting heart rate of 72 beats per minute and a mean lifespan of 80 years.

- (i) Suggest **two** reasons why humans can have a different lifespan from other mammals.

(2)

- (ii) State why cardiac output provides more information than the resting heart rate.

(1)

- (iii) An African giraffe has the longest stride of all land mammals and can run at a speed of up to 60 km hr^{-1} .

Describe how the nervous system affects cardiac output to enable the giraffe to run at this maximum speed.

(4)

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(Total for Question 4 = 12 marks)